

Task 1: [customer-behavior-analysis](#)

Customer Behavior Analysis Summary

The customer behavior dataset was analyzed using Python and Jupyter Notebook to understand customer purchasing trends and transaction patterns. Data cleaning and preprocessing were performed to handle missing values and improve data quality. Statistical analysis such as mean, median, mode, and missing value detection was carried out on important columns.

Three visualizations were created to better understand customer behavior:

- Bar chart for product category distribution
- Histogram for customer spending distribution
- Pie chart for payment method analysis

The analysis revealed that certain product categories are purchased more frequently than others, indicating strong customer preference patterns. Spending distribution showed that a small number of customers contribute significantly higher sales compared to average customers. Payment method analysis identified the most preferred transaction method among customers.

Actionable Insights for Alfido Tech

1. Focus marketing campaigns on the most popular product categories to increase customer engagement and sales.
2. Target high-spending customers with loyalty rewards, personalized offers, and premium services to improve retention.
3. Optimize payment options by promoting the most commonly used payment methods for smoother customer transactions.

```
[37]: #Mean
df.mean(numeric_only=True)

[37]: Customer ID      25017.632092
Product Price      254.742724
Quantity           3.004936
Total Purchase Amount  2725.385196
Customer Age       43.798276
Returns            0.508824
Age               43.798276
Churn              0.200520
dtype: float64

[38]: #Median
df.median(numeric_only=True)

[38]: Customer ID      25011.0
Product Price      255.0
Quantity           3.0
Total Purchase Amount  2725.0
Customer Age       44.0
Returns            1.0
Age               44.0
Churn              0.0
dtype: float64

[39]: #mode
df.mode().iloc[0]

[39]: Customer ID      48382.0
Purchase Date      2020-01-01 15:49:02
Product Category    Electronics
Product Price       290.0
Quantity            4.0
Total Purchase Amount  2533.0
Payment Method       Credit Card
Customer Age        47.0
Returns             1.0
Customer Name       Michael Johnson
Age                47.0
Churn              Not a
```